NLP Text Classification using DistilBERT

1. Objective

The objective of this assignment is to build a transformer-based text classification model that can automatically classify customer support tickets into predefined categories.

2. Dataset

The dataset used is **"Tobi-Bueck/customer-support-tickets"** from Hugging Face.

The dataset contains:

- subject: Short description of the issue
- body: Detailed explanation of the issue
- type: Category label (Problem, Incident, Request, Change)

These labels represent different types of support tickets.

3. Methodology

3.1 Data Loading

The dataset was loaded using the Hugging Face datasets library.

3.2 Data Preprocessing

- Records with missing labels were removed
- Subject and body fields were combined into a single text input
- Labels were converted from text to numerical format for model training

3.3 Tokenization

The text data was tokenized using the DistilBERT tokenizer:

- Padding applied for uniform length
- Truncation applied to limit sequence size

3.4 Model Selection

A pre-trained transformer model **DistilBERT (distilbert-base-uncased)** was used for classification.

3.5 Model Training

- Training performed using Hugging Face Trainer API
- Dataset subset used: 300 samples
- Batch size: 8
- Number of epochs: 1

3.6 Evaluation

The model performance was evaluated using a subset of the dataset.

3.7 Prediction

The trained model was tested on a sample input to predict the category of the support ticket.

4. Results

The model successfully learned to classify support tickets into categories such as Problem, Incident, Request, and Change.

Example:

Input:
"My system is not working"

Predicted Output:
Incident

5. Conclusion

The transformer-based DistilBERT model effectively classifies customer support tickets. Even with a small dataset, the model produces reasonable predictions. This demonstrates the effectiveness of pre-trained language models for text classification tasks.

6. Tools and Libraries Used

- Python
- Hugging Face Transformers
- Hugging Face Datasets
- PyTorch